

Performance Report – DataGrid Component

Test Environment

Device: Laptop
CPU: 11th Gen Intel Core i3-1115G4 @ 3.00GHz
RAM: 12 GB (4 GB + 8 GB extended)
OS: Windows
Browser: Microsoft Edge

Dataset Configuration

Total Rows: 50,000
Total Columns: 10
Row Height: 36px
Row Virtualization: Enabled
Column Virtualization: Enabled

Scrolling Performance (FPS)

Vertical scrolling maintained an average frame rate between 55–60 FPS. Minor frame drops were observed during rapid scrolling, but overall interaction remained smooth. Horizontal scrolling performance was not tested independently.

Interaction Latency

Column sorting operations completed within a single frame (<16ms). Cell edit commit interactions completed in under 100ms. Column resizing showed slight lag under continuous interaction.

Memory Usage

JavaScript heap usage ranged between approximately 1090 MB and 1270 MB during testing. Memory usage stabilized after repeated scrolling and interactions. No memory leaks were observed during the test session.

Stability Observations

During initial loading of the large dataset, the browser displayed a temporary “Page Unresponsive” warning. No crashes were observed, and the application recovered and continued functioning correctly after loading.

Overall Assessment

Overall performance is assessed as acceptable given the dataset size and hardware constraints. Virtualization effectively limits DOM nodes, allowing the DataGrid to remain usable at scale.